

Summary of Data

Day Ahead

- Shape: 32520 x 9
 - **Interval_end_UTC** —> Hourly record
 - **Market** —> 1 Value(DAY_AHEAD_HOURLY)
 - **Location** —> 1 Value(MINN.HUB)
 - **Location_type** —> 1 Value(hub)
 - **Imp** —> 7,552 (23%)
 - Mean —> 34.6993
 - SD —> 32.5090
 - Min —> -9.4
 - 25th —> 21.07
 - Med —> 27.49
 - 75th —> 38.42
 - Max —> 810.31
 - **Energy** —> 11,069 (34%)
 - Mean —> 36.3464
 - SD —> 33.0102
 - Min —> 7.9
 - 25th —> 23.2375
 - Med —> 29.1199
 - 75th —> 38.9999
 - Max —> 891.5899
 - **Congestion** —> 4577 (14%)
 - Mean —> -0.7486
 - SD —> 10.2758
 - Min —> -220.71
 - 25th —> -3.5
 - Med —> -0.16
 - 75th —> 2.36
 - Max —> 140.21
 - **loss** —> 1605 (5%)
 - Mean —> -0.8985
 - SD —> 2.3920
 - Min —> -73.45
 - 25th —> -1.61
 - Med —> -0.87
 - 75th —> -0.08
 - Max —> 18.3
 - **Year**
-

Load

Shape: 390,240 x 4

- **Interval_end_utc** —> Every 5-minute record
- **Publish_time_utc** —> 1345 (Daily publishing time?)
- **Load_forecast** —> 56767 (15%)
 - Mean —> 76895.1361
 - SD —> 12195.7770
 - Min —> 53925
 - 25th —> 68496
 - Med —> 74456
 - 75th —> 82797.25
 - Max —> 127807
- **Year**

Outage

Shape: 37,468 x 9

- **Interval_start_utc** —> 1355
- **Interval_end_utc** —> 1355
- **Publish_time_utc** —> 1344 ?????
- **Region** — 4
 - Central
 - South
 - North
 - MISO
- **derated_outages_mw** —> 8227(22%)
 - Mean —> 2963.0060
 - SD —> 2265.2516
 - Min —> 5
 - 25th —> 1134
 - Med —> 2377
 - 75th —> 4223
 - Max —> 14776
- **Forced_outage_mw** —> 7273 (19%)
 - Mean —> 2622.1974
 - SD —> 2281.1034
 - Min —> 33
 - 25th —> 977
 - Med —> 1935
 - 75th —> 3547
 - Max —> 18586
- **Planned_outages_mw** —> 11024 (29%)
 - Mean —> 9874.3918
 - SD —> 9189.3644
 - Min —> 193
 - 25th —> 3007
 - Med —> 7274
 - 75th —> 12832.25
 - Max —> 49272
- **unplanned_outages_mw** —> 8706 (23%)
 - Missing —> 218(<1%)
 - Mean —> 3041.8355
 - SD —> 2664.1069
 - Min —> 0
 - 25th —> 1017
 - Med —> 2110
 - 75th —> 4460
 - Max —> 18308
- **Year**

Real Time 5

- Shape: 360,723 x 9
- **Interval_end_UTC** —> Every 5-minute record
- **Market** —> 2 Values
 - Real_time_5_min (>99%)
 - Real_time_5_min_post_prelim (<1%)
- **Location** —> 1 Value(MINN.HUB)
- **Location_type** —> 1 Value(hub)
- **Imp** —> 21341 (6%)
 - Mean —> 34.4081
 - SD —> 83.79872
 - Min —> -161.25
 - 25th —> 18.65
 - Med —> 24.87
 - 75th —> 33.93
 - Max —> 10000
- **Energy** —> 34212 (9%)
 - Mean —> 35.9093
 - SD —> 82.1075
 - Min —> -37.15
 - 25th —> 21.12
 - Med —> 26
 - 75th —> 33.88
 - Max —> 100
- **Congestion** —> 17013 (5%)
 - Mean —> -0.8631
 - SD —> 22.1752
 - Min —> -604.52
 - 25th —> -3.06
 - Med —> 0
 - 75th —> 1.17
 - Max —> 574.02
- **Year**
- **Loss** —> 4464 (1%)
 - Mean —> -0.6381
 - SD —> 3.9682
 - Min —> -219.17
 - 25th —> -1.37
 - Med —> -0.71
- 75th —> 0.06
- Max —> 200.05

Solar

Shape: 224,638 x 8

- **Interval_start_utc** —> 32520
- **Interval_end_utc** —> 32520
- **Publish_time_utc** —> 1345 ????
- **North** —> 14402 (6%)
 - Mean —> 199.8974
 - SD —> 319.0402
 - Min —> 0.0
 - 25th —> 0.0
 - Med —> 7.2
 - 75th —> 295.5
 - Max —> 1700
- **Central** —> 56975 (25%)
 - Mean —> 1515.2339
 - SD —> 2561.4491
 - Min —> 0.0
 - 25th —> 0.0
 - Med —> 63.2
 - 75th —> 1880.275
 - Max —> 13144.2
- **South** —> 36795 (16%)
 - Mean —> 703.2500
 - SD —> 1133.7300
 - Min —> 0.0
 - 25th —> 0.0
 - Med —> 53
 - 75th —> 979.775
 - Max —> 5027.8
- **Miso** —> 70335 (31%)
 - Mean —> 2418.3814
 - SD —> 3935.4856
 - Min —> 0.0
 - 25th —> 0.0
 - Med —> 143.8
 - 75th —> 3147.275
 - Max —> 18869
- **Year**

Wind

Shape: 32520 x 8

- **Interval_start_utc** —> 32520
- **Interval_end_utc** —> 32520
- **Publish_time_utc** —> 1345 ????
- **North** —> 128118 (57%)
 - Mean —> 8685.9871
 - SD —> 4954.0581
 - Min —> 0.0
 - 25th —> 4501.275
 - Med —> 8050.4
 - 75th —> 12582.775
 - Max —> 20360.6
- **Central** —> 66188 (29%)
 - Mean —> 2761.1674
 - SD —> 1842.4317
 - Min —> 0.0
 - 25th —> 1258.1
 - Med —> 2369.05
 - 75th —> 3989.1749
 - Max —> 8296.3
- **South** —> 3127 (1%)
 - Mean —> 52.1582
 - SD —> 71.9816
 - Min —> 0.0
 - 25th —> 0.0
 - Med —> 17.7
 - 75th —> 81.0
 - Max —> 315.8
- **Miso** —> 70335 (31%)
 - Mean —> 2418.3814
 - SD —> 3935.4856
 - Min —> 0.0
 - 25th —> 0.0
 - Med —> 143.8
 - 75th —> 3147.275
 - Max —> 18869.0
- **Year**